

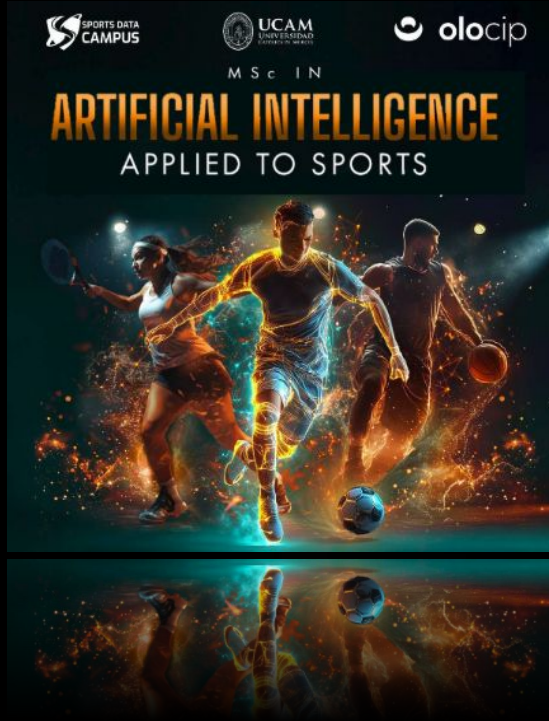
MASTER IN ARTIFICIAL INTELLIGENCE APPLIED TO SPORTS

Module 9

The great challenge: master's final project

**From theory to real-world
application**





Guidelines for the Master's Final Project in Artificial Intelligence Applied to Sports

Why is this project important?

- This is not just a final academic requirement
- It is the first time you face a real, open-ended problem
- There is no single correct solution
- It reflects how you think, structure, and solve problems



What makes this project unique?

- Combines theory and real-world application
- Requires both technical and analytical thinking
- Focuses not only on building models, but on solving problems
- Requires clear communication to different audiences



Project Objective

- Develop a complete AI solution applied to sports
- Demonstrate technical competence in data and modeling
- Structure the project using a professional methodology
- Communicate insights clearly and effectively



Thinking like a professional

You are not solving a
classroom exercise

You are solving a real
problem

Decisions should be
justified

Trade-offs must be
explained

Results must generate
value

What is expected from you?

- A complete and structured project
- A clear and impactful presentation
- A real-world application
- A balance between technical depth and communication



Project Deliverables

Documentation to Submit:

1. Final Project Report (Master's Thesis):

The final report must be structured into three levels, each tailored to a different audience, adapting the level of detail and language to both visual and oral formats.

2. Explanatory Video

A video that details each step of the CRISP-DM methodology implemented in the project. This video must be accessible and easy to understand, designed to educate the audience on how CRISP-DM was specifically applied within the context of the project.

- A structured final report
- An explanatory video
- Supporting materials (code, datasets, documentation)
- A clear narrative connecting all parts



Deadlines and Deliverables

Submission Dates:

The previously established deadlines, as outlined in the master's calendar, remain unchanged.

3-Level Report (PDF) & Explanatory Video:

Both must be submitted as part of the final project package.

Additional Files:

You must include all notebooks used, datasets, relevant papers, and a PDF with the bibliography.

Note:

If file sizes exceed platform limits, you may submit a text file containing links to cloud storage. Videos can also be uploaded as unlisted or private on YouTube.



CORE STRUCTURE

The 3-Level Communication Approach

Level 1 — Executive Summary

Content:

- Short and impactful overview of the project
- Focus on:
 - Problem
 - Solution
 - Impact
- No technical jargon
- Designed for decision-makers

Level 2 — Understanding the Big Picture

Content:

- What problem are you solving?
- Why is it relevant in sports?
- What are the expected benefits?
- Who is impacted by this solution?

Focus:

- Context
- Application
- Value

Level 3 — Technical Deep Dive

Content:

- Data sources and characteristics
- Data cleaning and preprocessing
- Modeling techniques used
- Evaluation metrics
- Final results

Key idea:

- Show rigor, not just results

Step 1: Business Understanding

Understanding the problem

- Define the problem clearly
- Identify stakeholders
- Understand the impact
- Translate business needs into data problems

Step 2–3: Data Understanding & Preparation

Working with data

- Explore the dataset
- Identify patterns and issues
- Handle missing values
- Transform and clean data
- Prepare features for modeling

Step 4–5: Modeling & Evaluation

Building and validating models

- Select appropriate algorithms
- Train and tune models
- Evaluate performance using metrics
- Compare different approaches

Step 6: Deployment

From model to real use

- How can this solution be used?
- Is it scalable?
- Can it be integrated into a real system?
- What is the business value?

Choosing the Right Project

How to choose your project?

- Choose a relevant problem
- Ensure data availability
- Keep scope realistic
- Focus on impact
- Align with your interests

From Idea to Structured Project

From Idea to Impact

- A good project is not just about the idea
- It is about how you develop it
- It must be:
 - Structured
 - Clear
 - Impactful



Key Takeaways

- Structure is as important as content
- Communication is as important as modeling
- Simplicity beats unnecessary complexity
- Impact matters more than perfection

The Key Requirement

**Anyone can have an idea.
What makes the difference is
how you turn that idea into a
real solution.**

- Solve a real problem
- Use a structured methodology
- Communicate effectively
- Adapt your message to different audiences

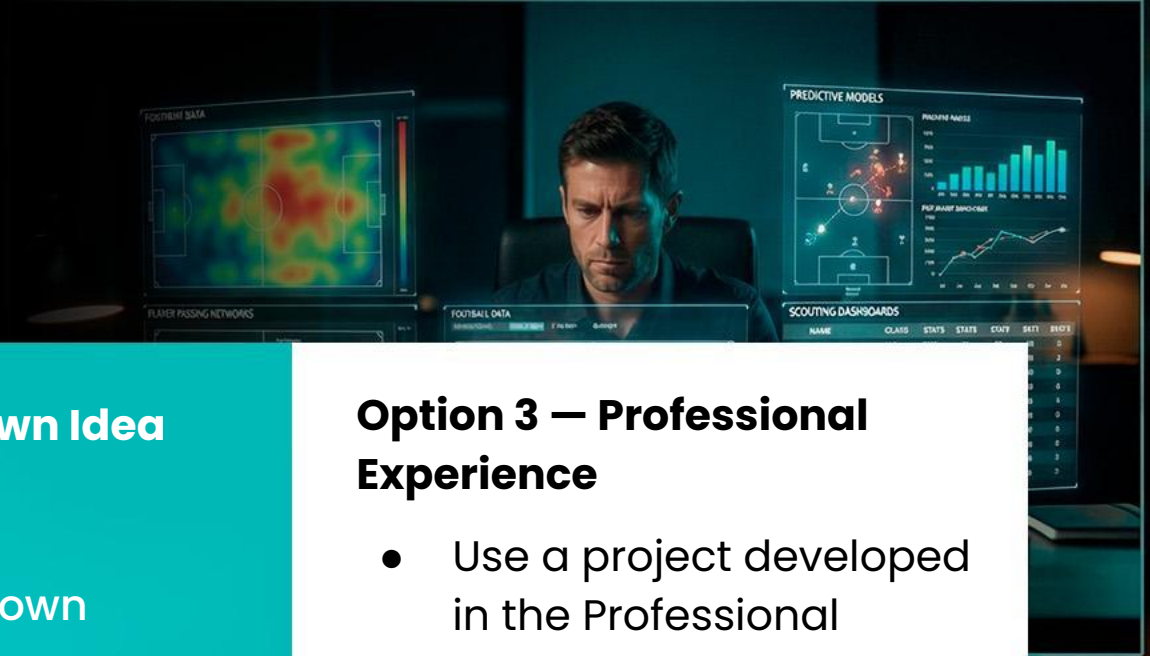
Common Mistakes

What to avoid

- Focusing only on the model
- Ignoring the problem definition
- Poor communication
- Lack of structure
- No real-world perspective



Choosing Your Final Project



Option 1 — Proposed Topics

- Choose from the predefined project ideas
- Already aligned with real industry problems
- Clear scope and direction

Option 2 — Your Own Idea

- Propose your own project topic
- Focus on innovation and creativity
- Explore areas aligned with your interests
- Must be relevant and feasible

Option 3 — Professional Experience

- Use a project developed in the Professional Extension Department
- Work on a real case with real data
- Apply your knowledge in a practical environment

AI Project Areas

The key is not choosing the most complex area — but choosing the one where you can create the most impact.

Machine Learning (Foundations)

- Predictive models (classification, regression)
- Clustering and pattern discovery
- Time series analysis
- Applications: performance prediction, match outcomes

Deep Learning

- Neural networks for complex patterns
- CNNs (images), RNNs/Transformers (sequences)
- Applications across text, image, and video

Computer Vision (Images & Video)

- Image classification
- Object detection & tracking
- Video analysis
- Applications: player tracking, tactical analysis

Natural Language Processing

- Sentiment analysis
- Text classification
- Topic modeling
- Applications: fan opinions, media analysis, reports



AI Project Areas

The key is not choosing the most complex area — but choosing the one where you can create the most impact.

Generative AI

- Text generation (LLMs)
- Image generation
- Data augmentation
- Applications: reports, simulations, content creation

AI Agents & Systems

- Autonomous decision-making systems
- Tool-using AI (APIs, retrieval, reasoning)
- Applications: assistants, analysis platforms

Multimodal AI

- Combining text, image, audio, and video
- Understanding complex real-world scenarios
- Applications: full match analysis systems

Audio Processing

- Speech recognition
- Emotion detection
- Audio classification
- Applications: commentary analysis, player communication



Course Participants Grades Reports

I MSc in IA Applied to Sports

General information



WhatsApp Group

[Follow the link and join the group](#)

Official communications

 Notice board

This board will serve to make the most of the communications of the master's degree from the management and coordination. You should always be very attentive to all the messages that arrive from this forum because they will contain information of interest. This forum does not allow replies

 Course Forum

Through this forum we will manage the common aspects of the Master: presentations, your contributions and everything you want to share.
However, one important thing, so that your doubts are well registered, do not transfer them through this forum, but from the one that is enabled for this purpose, in each of the modules. 😊

Submit a Document with:

- Description
- Objectives
- Data Sources
- Technologies/tools/programming languages
- Expected Value - Outcome





**This project is where
everything comes together.
Not just what you know — but
what you can do with it.**

We wish you all the best in this
journey.

We will be here to support you
every step of the way.



Thank you!

ARTIFICIAL INTELLIGENCE APPLIED TO SPORTS



PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

In sports technology projects, we face the constant challenge of combining technical rigor with accessibility for non-technical professionals. In this context, the structure of professional project documentation is carefully designed to serve as a comprehensive tool that guides students in presenting their projects, ensuring that every aspect of the work is effectively communicated to a wide range of audiences, from executives to data experts.

The structure proposed below is intended to serve as a basic framework for professional projects, providing a clear roadmap for students. At the same time, the need for adaptability is recognized.

We suggest organizing the table of contents in the following way, so that each section is aligned with the appropriate level of detail for different types of readers.

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

Table of Contents

1. Executive Summary (Level 1: Executive Perspective)

This section is the entry point to your project, especially for non-technical stakeholders such as managers, decision-makers, or business professionals.

It should provide a clear, concise, and impactful overview of your work, answering the most important questions:

- What problem are you solving?
- Why is it relevant?
- What solution did you develop?
- What results did you achieve?
- What is the impact of your work?

The key here is simplicity and clarity. Avoid technical jargon and focus on communicating value and relevance.

If someone only reads this section, they should fully understand why your project matters.

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

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2. Introduction (Level 2: Macro Vision)

In this section, you provide the broader context of your project.

You should:

- clearly define the problem
- explain why it is relevant in the sports industry
- describe the current situation and challenges
- identify who is affected by this problem

You also introduce:

- the objectives of the project
- the expected benefits (technical, strategic, or operational)

Why does this project exist, and why should anyone care?

It connects your work to the real world, showing that your project is not just technical, but meaningful.

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

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3. Objectives

Here you define what your project is trying to achieve in a structured way.

You should:

- break down the main goal into smaller, specific objectives
- ensure they are measurable and realistic
- align them with the problem you introduced

Good objectives are:

- clear (no ambiguity)
- focused (not too broad)
- achievable within your scope

Avoid vague statements like: Improve performance

Instead: Develop a model to predict player performance with X accuracy using historical data

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

Table of Contents

4. Timeline (Project Planning)

This section shows your ability to plan and manage a project professionally.

You should include:

- a timeline of the project phases (Gantt chart is recommended)
- key milestones and deliverables
- dependencies between tasks

You should also consider:

- constraints (data availability, time limitations)
- business rules that may affect execution

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

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5. Conceptual and Technological Architecture of the Project

Here you describe how your system is designed at a high level.

You should explain:

- the overall architecture (how everything fits together)
- the tools and technologies used (Python, APIs, frameworks, etc.)
- how data flows through the system
- how different components interact

This is where you show:

- system design thinking
- understanding of infrastructure
- ability to justify technical decisions

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

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6. Methodologies and Techniques Used (Level 3: Technical Detail)

In this section, you explain the methodology guiding your project.

The focus is on CRISP-DM, which provides a structured approach to solving data problems.

You should:

- describe each phase of CRISP-DM
- explain why this methodology is appropriate
- show how it helps structure your work

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

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7. Project Development (Following CRISP-DM Phases)

This is the core of your project, where you explain everything you did in detail.

1. Business Understanding

- Define the real-world problem clearly.
- Explain the objectives from a business or application perspective.

2. Data Understanding

Describe the data you are using:

- sources
- structure
- limitations
- initial insights

3. Data Preparation

Explain how you prepared the data:

- cleaning
- handling missing values
- transformations
- feature engineering

4. Modeling

Describe:

- models used
- why you chose them
- how you trained them

5. Evaluation

Explain:

- performance metrics
- results
- comparison between models
- limitations

6. Deployment

Describe how your solution could be used in practice:

- real-world application
- scalability
- integration possibilities

Table of Contents

8. Results Discussion

This section goes beyond presenting results — it is about interpreting them.

You should:

- explain what the results mean
- assess whether they are useful in practice
- identify strengths and limitations

Avoid just reporting metrics.

Focus on insight and understanding.

Table of Contents

9. Conclusions and Future Work (Level 3: Technical Detail)

Here you bring everything together.

You should:

- summarize key findings
- reflect on what you achieved
- connect results back to objectives

Then, propose:

- improvements
- extensions
- future applications

vision | ability to reflect | long-term thinking

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

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10. Bibliography

include all references used:

- academic papers
- datasets
- tools
- external sources

11. Appendices

This section includes additional material that supports your project but is not essential in the main document.

Examples:

- code snippets
- extended results
- technical documentation
- datasets

This allows you to:

- keep your main document clean
- still provide full technical depth

PROPOSED DOCUMENTATION STRUCTURE FOR PROFESSIONAL PROJECTS

This structure ensures that each section of the work is relevant to different types of audiences.

The Executive Summary is designed to capture the attention of non-technical executives, while the Methodology and Project Development sections provide the level of detail that technical experts and specialists expect.

The Introduction and Conclusions serve both audiences, offering both a general overview and a detailed closure of the project, as well as its potential future directions.

Final Note on Structural Flexibility

It is important to emphasize that the structure presented here is only a proposal and can be adapted according to the specific needs of each project.

Students are encouraged to customize this structure so that it better aligns with the unique objectives and requirements of their individual research.

This flexibility is essential to foster a creative and personalized approach, encouraging students to take full ownership of their work and present it in a way that resonates with both technical experts and industry thought leaders.